

Dr. DEEPSHIKHA BHATI

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PROFILE SUMMARY:

Academic and research professional with over nine years of experience in teaching and research, specializing in Explainable Artificial Intelligence (XAI), Large Language Models (LLMs), Generative AI, Information Visualization, Visual Analytics, Human–Computer Interaction, Deep Learning, Machine Learning, and Image Processing. Research focuses on developing interpretable and human-centered AI systems, with applications in medical imaging, AI-assisted document reading, educational technologies, and visual analytics. Actively engaged in collaborative and interdisciplinary research, with a broader goal of advancing trustworthy AI through explainability, visualization, and interactive intelligence.

RESEARCH INTERESTS:

Explainable Artificial Intelligence (XAI); Information Visualization and Visual Analytics; Large Language Models (LLMs) and Generative AI; Human–Computer Interaction; Medical Imaging; Educational AI.

Google Scholar: 923 citations — <https://scholar.google.com/citations?user=FrODSEAAAAAJ&hl=en>

EDUCATION:

Ph.D. in Computer Science, Kent State University, Kent, Ohio, USA, 2026.

Dissertation: Towards Enhanced AI Explanation and Interaction with Users; Advisor: Prof. Ye Zhao

Master of Technology in Computer Science, Dr. A.P.J. Abdul Kalam Technical University, India, 2017.

Bachelor of Technology in Computer Science, Dr. A.P.J. Abdul Kalam Technical University, India, 2014.

INDUSTRIAL EMPLOYMENT AND EXPERIENCE:

Kent Displays, Inc. | Makers of Boogie Board, Kent, Ohio, USA

Mobile Application Developer, May 2021–August 2021 (Full Time), August 2021–December 2021 (Part-Time)

Link: <https://play.google.com/store/apps/details?id=com.kentdisplays.blackboard>

Blackboard™ by Boogie Board™ is an electronic writing tool to plan, sketch, and design featuring Liquid Crystal Paper™. The free Blackboard app makes it easy to save, organize, search, and share your work, doodles, and designs and includes the following features:

- **Capture your work:** The Blackboard app uses your mobile device camera to scan your Blackboard.
- **Carbon copy compatible:** With Boogie Board's Carbon Copy Pen and Smart Template, you can digitally save all your creations as you write and stream them to your mobile device with this app. You will never lose a great idea again!
- **Edit and Enhance:** The Blackboard app includes tools to draw, highlight, and enhance your work.
- **Share:** The Blackboard app archives your work as images or PDFs and makes sharing to emails, messages, and more a cinch.

BOOKS (SPRINGER NATURE)

1. Published Book

Title: *A Beginner's Guide to Generative AI: An Introductory Path to Diffusion Models, ChatGPT, and LLMs*
Authors: Deepshikha Bhati, FNU Neha, Angela Guercio, Md Amiruzzaman, and Aloysius Bhati Kasturiarachi
Publisher: Springer Nature, Synthesis Lectures on Computer Science
Link: <https://link.springer.com/book/10.1007/978-3-031-84724-0>

2. Book Under Contract

Title: *The Journey of Large Language Models and Retrieval-Augmented Generation Systems: Architectures and Innovations Behind GPT, LLaMA, Gemini, Claude, Falcon, DeepSeek, Qwen, Grok, and RAG Variants*
Authors: Deepshikha Bhati, FNU Neha, Aloysius Bhati Kasturiarachi, and Angela Guercio
Publisher: Springer Nature, Synthesis Lectures on Computer Science
Status: Under contract; publication in progress

GRANTS (FUNDED)

Summer Teaching Development Grant (UTC)

Title: *EasyLearn-Canvas: An AI-Augmented Metacognitive Feedback and Quiz Generation System Integrated with Canvas LMS*
Role: Principal Investigator
Funding Agency: University Teaching Council (UTC), Kent State University
Amount: \$9,000
Date: April 2026 – January 2027
Description: Development of an AI-enhanced system integrating retrieval-augmented feedback and automated quiz generation within Canvas LMS to support metacognitive learning and improve student outcomes.

GRANT PROPOSALS

Spencer Foundation Vision Grant Proposal (2025)

Title: *EquiLearn: AI-Augmented Metacognitive Feedback System for Equitable Test Preparation*
Role: Co-Principal Investigator
Funding Agency: Spencer Foundation (Vision Grant Program)
Amount Requested: \$75,000
Submitted: September 2025
Status: Not Funded (February 2026)

AWS Agentic AI Proposal (2026)

Title: *Agentic AI for Equitable Learning: Integrating Multi-Agent Feedback and Multimodal Quiz Generation*

Role: Principal Investigator

Funding Agency: Amazon Web Services (AWS Agentic AI Call for Proposals)

Amount Requested: \$70,000

Status: Under Review, 2026

Description: Proposed *EduLearn-Agentic*, an open-source agentic AI system combining multimodal lecture ingestion, quiz generation, and retrieval-augmented metacognitive feedback using AWS Bedrock, SageMaker, and FAISS-based vector memory.

PUBLICATIONS:

Refereed Journal Publications (Published)

1. **Bhati D.**, Zhao, Y., Wu, T. H., Amiruzzaman, M., and Yang, J. (2026). Easy Interpretation of Image Classification Results with Feature-level Visualization. *Electronic Imaging*, 38, 1-13.
Link:<https://library.imaging.org/ei/articles/38/1/VDA-349>
2. **Bhati, D.**, Neha, F., Bandaru, D. S., Weber, M., and Gajera, I. D. (2026). Mapping the LLM Landscape: A Cross-Family Survey of Architectures, Alignment Methods, and Benchmark Performance. *AI*, 7(4), 142.
Link:<https://www.mdpi.com/2673-2688/7/4/142>
3. Neha, F., **Bhati, D.**, and Shukla, D. K., "Phenotyping of Histology Imaging Data with Histomics," *AI*, vol. 7, no. 6, article 228, 2026.
Link: <https://www.mdpi.com/2673-2688/7/6/228>
4. **Bhati D.**, Amiruzzaman M., Zhao Y., Guercio A., and Le T., "A Survey of Post-Hoc XAI Methods from a Visualization Perspective: Challenges and Opportunities," *IEEE Access* 2025.
Link:<https://ieeexplore.ieee.org/document/11039632>
5. Neha, F., **Bhati D.**, Shukla, D.K., Dalvi, S.M., Mantzou, N. and Shubbar, S., "U-Net in Medical Image Segmentation: A Review of Its Applications Across Modalities." *Healthcare Analytics*, 100416, Elsevier, 2025.
Link:<https://www.sciencedirect.com/science/article/pii/S2772442525000358>
6. Neha, F., **Bhati D.**, "Retrieval-Augmented Generation (RAG) in Healthcare: A Comprehensive Review." *AI* 6.9 (2025): 226.
Link:<https://doi.org/10.3390/ai6090226>
7. Neha, F., **Bhati, D.** and Shukla, D.K., 2025, "Generative AI Models (2018–2024): Advancements and Applications in Kidney Care", *BioMedInformatics*. 2025; 5(2):18.
Link:<https://doi.org/10.3390/biomedinformatics5020018>
8. Amiruzzaman, S., Amiruzzaman, M., Begum, H., **Bhati, D.** and Wu, T.H., "Deepening Mathematical Understanding Using Visualization and Interactive Learning for Deaf Students" in *Engineering Proceedings* 89, no. 1 (2025): 4
Link:<https://www.mdpi.com/2673-4591/89/1/4>
9. Le, T., Ibrahim N., Nguyen T., Noiplab T., Kim J., and **Bhati D.**, "Enhanced Comparative Analysis of Pretrained and Custom Deep Convolutional Neural Networks for Galaxy Morphology Classification," in *Engineering Proceedings* 89, no. 1 (2025): 36.
Link:<https://www.mdpi.com/2673-4591/89/1/36>

10. Wu T. H., Amiruzzaman M., Zhao Y., **Bhati D.** and Yang J., "Visualizing Routes With AI-Discovered Street-View Patterns," in *IEEE Transactions on Computational Social Systems*, vol. 11, no. 5, pp. 6380-6391, Oct. 2024.
Link:<https://ieeexplore.ieee.org/document/10504858>
11. **Bhati D.**, Neha, F. and Amiruzzaman, M., "A survey on explainable artificial intelligence (xai) techniques for visualizing deep learning models in medical imaging." *Journal of Imaging* 10, no. 10 (2024): 239.
Link:<https://www.mdpi.com/2313-433X/10/10/239>
12. Neha, F., **Bhati D.**, Shukla, D.K. and Amiruzzaman, M., "Chatgpt: Transforming healthcare with AI." *AI* 5, no. 4 (2024): 2618-2650.
Link:<https://www.mdpi.com/2673-2688/5/4/126>
13. Kumar, J.S., Amiruzzaman, M., Bhuiyan, A. A. and **Bhati D.**, "Predictive Analytics in Law Enforcement: Unveiling Patterns in NYPD Crime through Machine Learning and Data Mining." *Research Briefs on Information and Communication Technology Evolution* 10 (2024): 36-59.
Link:<https://rebicte.org/index.php/rebicte/article/view/188>

Journal Papers Under Review

1. **Bhati, D.**, Zhao, Y., Guercio, A., and Wu, T. H., "Bridging Retrieval and Interaction: A Survey of AI-Powered Document Reading Systems," *ACM Transactions on Interactive Intelligent Systems (TiiS)*, under review, 2026.
2. **Bhati, D.**, Guercio, A., Le, T., Bandaru, D. S., Neha, F., Auman, C., and Rahaei, A., "EasyLearn: A Retrieval-Augmented Learning System for Grounded Feedback and Metacognitive Support in STEM Education," *ACM Transactions on Computing Education (TOCE)*, under review, 2026.

Refereed Conference Publications:

1. **Bhati, D.**, Le, T., Guercio, A., Auman, C., and Bandaru, D. S., "Retrieval-Augmented AI Feedback to Support Metacognition and Accessible Feedback in STEM Education," in *2026 IEEE Conference on Artificial Intelligence (CAI)*, Granada, Spain, May 8–10, 2026, pp. 1258–1263, IEEE.
DOI: <https://doi.org/10.1109/CAI68641.2026.11536256>
2. Auman, C., **Bhati, D.**, Guercio, A., and Neha, F., "Interpreting Text-to-Image Diffusion Through Structured Prompts and Cross-Attention Attribution," in *2026 IEEE Conference on Artificial Intelligence (CAI)*, Granada, Spain, May 8–10, 2026, pp. 1245–1251, IEEE.
DOI: <https://doi.org/10.1109/CAI68641.2026.11536521>
3. Auman, C., **Bhati, D.**, Arquilla, K., Neha, F., and Guercio, A., 2025, "Integrating Layer-Wise Relevance Propagation with Stable Diffusion for Enhanced Interpretability", in *EuroVis Workshop on Visual Analytics (EuroVA)*.
Link: <https://diglib.eg.org/handle/10.2312/eurova20251102>
4. Craver, N., **Bhati, D.** and Guercio, A., 2025, "Integrating LLM Text-to-Speech for an Accessible Bible Study Application", in *2025 IEEE World AI IoT Congress (AIIoT)*, Seattle, WA, USA, 2025, pp. 0912-0918
Link: <https://ieeexplore.ieee.org/document/11105356>
5. Neha, F., **Bhati, D.**, Shukla, D. K., and Bhati, D. , "Evaluating the multimodal capabilities of ChatGPT for kidney disease diagnosis: integrating text and medical images " in *International Conference on AI-Generated Content (AIGC 2024)* (Vol. 13649, pp. 184-191). SPIE.
Link: <https://doi.org/10.1117/12.3065777>

6. Neha, F., **Bhati D.**, "DeepSeek Models in STEM Education: Capabilities, Applications, and Challenges", 2025 7th International Workshop on Artificial Intelligence and Education (WAIE), September 27-29, 2025, Yokohama, Japan (Accepted)
Link: <https://www.techrxiv.org/doi/full/10.36227/techrxiv.174198447.71647707>
7. Neha, F. and **Bhati D.**, 2025," Traditional RAG vs. Agentic RAG: A Comparative Study of Retrieval-Augmented Systems", in IEEE International Conference on Future Machine Learning and Data Science, FMLDS 2025 (Accepted)
Link: <https://www.techrxiv.org/doi/full/10.36227/techrxiv.175624551.12254549>
8. Neha, F. and **Bhati D.**, 2025," A Survey of DeepSeek Models", in ISCMi 2025, 12th IEEE International Conference on Soft Computing & Machine Intelligence (Accepted)
Link: <https://www.techrxiv.org/doi/full/10.36227/techrxiv.173896582.25938392>
9. Neha, F., **Bhati D.**, Shukla, D.K. and Amiruzzaman, M., " A Tiered GAN Approach for Monet-Style Image Generation.", 34th IEEE International Conference on Computer Theory and Applications (ICCTA 2024), 14th-16th December 2024, Alex, Egypt (Accepted)
10. Neha, F., **Bhati D.**, Shukla, D.K. and Amiruzzaman, M., "From classical techniques to convolution-based models: A review of object detection algorithms."In 2025 IEEE 6th International Conference on Image Processing, Applications and Systems (IPAS) (pp. 1-6). IEEE.
Link: <https://ieeexplore.ieee.org/document/10924494>
11. Neha, F.,**Bhati D.**, Shukla D. K., Guercio A., and Ward B., "Exploring AI Text Generation, Retrieval-Augmented Generation, and Detection Technologies: A Comprehensive Overview," 2025 IEEE 15th Annual Computing and Communication Workshop and Conference (CCWC), 6th - 8th January 2025, University of Nevada, Las Vegas, USA (pp. 00633-00639). IEEE
Link:<https://ieeexplore.ieee.org/abstract/document/10903902>
12. Ward B.,**Bhati D.**, Neha F., and Guercio A., "Analyzing the Impact of AI Tools on Student Study Habits and Academic Performance," 2025 IEEE 15th Annual Computing and Communication Workshop and Conference (CCWC), 6th - 8th January 2025, University of Nevada, Las Vegas, USA. (pp. 00434-00440). IEEE.
Link:<https://ieeexplore.ieee.org/abstract/document/10903692>
13. **Bhati D.**, Neha F., Amiruzzaman M., Guercio A., Shukla D. K., and Ward B., "Neural Network Interpretability with Layer-Wise Relevance Propagation: Novel Techniques for Neuron Selection and Visualization," 2025 IEEE 15th Annual Computing and Communication Workshop and Conference (CCWC), 6th - 8th January 2025, University of Nevada, Las Vegas, USA. (pp. 00441-00447). IEEE. **Link:**<https://ieeexplore.ieee.org/abstract/document/10903721>
14. Arquilla, K., Gajera I., Darling M., **Bhati D.**, Singh A., and Guercio A., "Exploring Fine-Grained Feature Analysis for Bird Species Classification using Layer-wise Relevance Propagation." In 2024 IEEE World AI IoT Congress (AIIoT), pp. 625-631. IEEE, 2024.
Link:<https://ieeexplore.ieee.org/document/10579007>
15. Aditi S., Kumar S., Ehtesham A., Khoei T. T., and **Bhati D.**, "Large Language Model-Driven Immersive Agent." In 2024 IEEE World AI IoT Congress (AIIoT), pp. 0619-0624. IEEE, 2024
Link:<https://ieeexplore.ieee.org/document/10578948>
16. **Bhati D.**, Guercio A., Rossano V., and Francese R., "BookMate: Leveraging Deep Learning to Empower Caregivers of People with ASD in Generation of Social Stories." In 2023 27th International Conference Information

Visualisation (IV), pp. 403-408. IEEE, 2023.

Link:<https://ieeexplore.ieee.org/abstract/document/10303403>

17. Jamonnak, S., **Bhati D.**, Amiruzzaman, M., Zhao, Y., Ye, X. and Curtis, A., 2022. VisualCommunity: a platform for archiving and studying communities. Journal of computational social science, 5(2), pp.1257-1279.
Link:<https://link.springer.com/article/10.1007/s42001-022-00170-y>
18. Francese, R., Guercio, A., Rossano, V. and **Bhati D.**, 2022, June. A Multimodal Conversational Interface to Support the creation of customized Social Stories for People with ASD. In Proceedings of the 2022 International Conference on Advanced Visual Interfaces (pp. 1-5).
Link:<https://dl.acm.org/doi/10.1145/3531073.3531118>
19. **Bhati D.**, Amiruzzaman, M., Jamonnak, S. and Zhao, Y., 2021, December. Interactive visualization and capture of geo-coded multimedia data on mobile devices. In International Conference on Intelligent Human Computer Interaction (pp. 260-271). Cham: Springer International Publishing.
Link:https://link.springer.com/chapter/10.1007/978-3-030-98404-5_25
20. **Bhati D.**, 2017. Face Recognition Stationed on DT-CWT and Improved 2DPCA employing SVM Classifier. International Journal of Computer Applications, 975, p.8887.
Links:<https://www.ijcaonline.org/archives/volume159/number5/bhati-2017-ijca-912944.pdf>
21. Pal, M., **Bhati D.**, Kaushik, B. and Banka, H., 2015. Solving Classification Problem using Reduced Dimension and Eigen Structure in RSVM. International Journal of Computer Applications, 975, p.8887
Links:<https://ijcaonline.org/archives/volume117/number21/20682-3537/>
22. **Bhati D.** and Gupta, V., 2015. Survey—A comparative analysis of face recognition technique. Int. J. Eng. Res. General Sci, 3(2), pp.597-609
Links:<http://pnrsolution.org/Datacenter/Vol3/Issue2/82.pdf>

Book Chapter Publications

1. Srivastava A., Trivedi M. C., Saha A., Saluja S., and **Bhati D.**, "Current Advances in Locality-Based and Feature-Based Transformers: A Review." In International Conference on Data & Information Sciences, pp. 321-335. Singapore: Springer Nature Singapore, 2023.
Link:https://link.springer.com/chapter/10.1007/978-981-99-6906-7_28
2. Joshi R., Trivedi M. C., Goyal V., and **Bhati D.**, "Recent Trends for Practicing Steganography Using Audio as Carrier: A Study." In Advances in Data and Information Sciences: Proceedings of ICDIS 2022, pp. 549-555. Singapore: Springer Nature Singapore, 2022.
Links:https://link.springer.com/chapter/10.1007/978-981-19-5292-0_52
3. Joshi R., Trivedi M. C., Goyal V., and **Bhati D.**, "DNA sequence in cryptography: A study." In Advances in Data and Information Sciences: Proceedings of ICDIS 2022, pp. 557-563. Singapore: Springer Nature Singapore, 2022.
Links:https://link.springer.com/chapter/10.1007/978-981-19-5292-0_53

ACADEMIC EMPLOYMENT AND EXPERIENCE:

1. *08.2026 to Present: Assistant Professor, Department of Computer Science, Kent State University at Stark.*

Fall 2026: CS III Programming Patterns (CS 44001), Introduction to Database System Design (CS 33007), CS II Data Structures and Abstraction (CS 23001), CS I Programming and Problem Solving (CS 13001)

2. 08.2022 to 08.2026: Full-Time Non-Tenure Track Lecturer, Department of Computer Science, Kent State University at Stark.

Spring 2026: Structure of Programming Languages (CS 33101), Human Interface Computing (CS 32301), Computer Science Principles (CS 10051), CS II Data Structures and Abstraction (CS 23001), CS I Programming and Problem Solving (CS 13001)

Fall 2025: CS III Programming Patterns (CS 44001), Introduction to Database System Design (CS 33007), Computer Science Principles (CS 10051), CS II Data Structures and Abstraction (CS 23001), CS I Programming and Problem Solving (CS 13001)

Spring 2025: Structure Of Programming Languages (CS 33101), Human Interface Computing (CS 32301), Design and Analysis of Algorithms (CS 46101), Algorithms and Programming I (CS 61002, Online Asynchronous)

Fall 2024: CS III Programming Patterns (CS 44001), Introduction to Database System Design (CS 33007), Algorithms and Programming I (CS 61002, Online Asynchronous), Algorithms and Programming II (CS 61003, Online Asynchronous)

Summer 2024: Computer Science Principles (CS 10051), Algorithms and Programming I (CS 61002, Online Asynchronous)

Spring 2024: Structure Of Programming Languages (CS 33101), Human Interface Computing (CS 32301), Computer Science Principles (CS 10051), CS I Programming and Problem Solving (CS 13001), Internship in Computer Science (CS 33192)

Fall 2023: CS III Programming Patterns (CS 44001), Introduction to Database System Design (CS 33007), CS II Data Structures and Abstraction (CS 23001), CS I Programming and Problem Solving (CS 13001), IT - Visual Basic Programming (CS 11002)

Summer 2023: CS II Data Structures and Abstraction (CS 23001)

Spring 2023: Structure Of Programming Languages (CS 33101), Human Interface Computing (CS 32301), Computer Science Principles (CS 10051), CS I Programming and Problem Solving (CS 13001)

Fall 2022: CS III Programming Patterns (CS 44001), Introduction to Database System Design (CS 33007), CS II Data Structures and Abstraction (CS 23001), CS I Programming and Problem Solving (CS 13001)

3. 08.2021 to 05.2022: Part-Time Instructor, Department of Computer Science, Kent State University.

Spring 2022 (Online Synchronous Course): Computer Science Principles (CS 10051), Algorithms and Programming II (CS 61003)

Fall 2021 (Online Synchronous Course): Computer Science Principles (CS 10051), FYE (First-Year Experience) (CS 10097)

4. ***08.2018 to 05.2021: Graduate Assistantship, Department of Computer Science, Kent State University.***

Spring 2021 (Online Synchronous Course): Computer Science Principles (CS 10051)

Fall 2020 (Online Synchronous Course): Computer Science Principles (CS 10051)

Spring 2020: Computer Science Principles (CS 10051)

Fall 2019: Computer Science Principles (CS 10051)

Spring 2019: Computer Science Principles (CS 10051), CS III Programming Patterns (CS 44001)

Fall 2018: Computer Science Principles (CS 10051)

5. ***08.2017 to 05.2018: Teaching Assistantship, Department of Computer Science, Kent State University.***

Lab Instructor and Grader for: *Advanced Digital Design, Systems Programming, Software Systems Requirements and Interaction Design, Drone Programming*

6. ***06.2022 to 08.2022, 06.2021 to 08.2021, and 06.2020 to 08.2020: Summer Part-Time Instructor, Department of Computer Science, Kent State University.***

CS II Data Structures and Abstraction (CS 23001)

7. ***06.2019 to 08.2019 and 06.2018 to 08.2018: Summer Software Developer, Department of Computer Science, Kent State University.***

Project: “Geo Visuals Mobile Application (GVM app)” developed and tested by Domain Experts

iPhone: <https://apps.apple.com/app/id1492419929>

Android: <https://play.google.com/store/apps/details?id=com.gvm.geovideo>

COURSE DEVELOPMENT:

Algorithms and Programming II (CS 61003, Online Asynchronous)

Kent State University, KSU Teach CS Program

August 2024

- Developed a comprehensive online asynchronous course on algorithms and programming to support the **CS Teach Endorsement Program** as part of ***KSU Teach CS State Grant: Innovative Methods for Preparing a New Generation of CS Teachers and Students in Ohio.***
- Designed course content to provide in-depth instruction in advanced algorithms, data structures, and programming methodologies to equip educators with practical skills and modern pedagogical strategies.
- Implemented interactive materials, assessments, and resources to promote self-paced learning, aligning with the program’s goals of enhancing computer science instruction across Ohio.

- Developed a resource-rich course specifically created to foster a stronger sense of belonging in an asynchronous environment.

SCHOLARSHIPS / AWARDS:

1. **Platinum Teaching Recognition Award (2026):** This award recognizes outstanding contributions to teaching innovation and student learning through the development of AI-augmented, retrieval-based feedback systems that support metacognitive engagement and accessible learning in STEM education.
2. **Kostas Pantazos Memorial Award for Outstanding Paper in Visualization and Data Analysis (2026):** Awarded for the paper “*Easy Interpretation of Image Classification Results with Feature-Level Visualization*,” *Electronic Imaging*, 38 (2026): 1–13.
3. **Google Certificate in Introduction to AI (2026):** Successfully completed an online course authorized by Google and offered through Coursera, covering foundational concepts in artificial intelligence, including machine learning, data-driven decision-making, and real-world AI applications. *(Issued: Feb 2026)*
4. **John and Fonda Elliot Design Innovation Faculty Fellowship (Summer 2025):** Selected as a Summer 2025 Faculty Fellow to participate in a six-week interdisciplinary fellowship supported by an endowed gift. Engaged in collaborative, human-centered design projects, co-teaching experiences, and innovation strategy development. Received \$5,000 stipend and certification as a Design Innovation Change Agent.
5. **Teaching Scholars Program Fellow (2025–2026):** Selected as a Teaching Scholar at Kent State University to design, implement, and assess a year-long scholarly teaching project. Participated in collaborative retreats and cohort-based mentoring to explore innovative strategies for enhancing student learning outcomes.
6. **Bronze Teaching Recognition Award (2024):** This award acknowledges my contributions to fostering belonging and student engagement in asynchronous environments
7. **Nominated for Distinguished Teaching Award, Kent State University, Stark in 2023, 2024, 2025 and 2026**
8. **IEEE International Conference on Knowledge Innovation and Invention (ICKII), Best Conference Paper Award** “Deepening Mathematical Understanding Using Visualization and Interactive Learning for Deaf Students” in **7th International Conference on Knowledge Innovation and Invention 2024**
9. **IEEE International Conference on Knowledge Innovation and Invention (ICKII), Best Conference Paper Award** “Enhanced Comparative Analysis of Pretrained and Custom Deep Convolutional Neural Networks for Galaxy Morphology Classification” in **7th International Conference on Knowledge Innovation and Invention 2024**
10. **Summer 2024 - Belonging Changemakers Summer Project Program.** The project was selected to create opportunities for student engagement using Belonging principles in asynchronous courses and received a stipend upon completion.
11. **IHCI-2021, Best Paper Session Award** “Interactive Visualization and Capturing Geo-Coded Multimedia Data on Mobile Devices” in **13th International Conference on Intelligent Human-Computer Interaction**
12. **Postmatric Scholarship** (Master in Computer Science and Technology Degree) for OBC, Uttar Pradesh 2014 to 2016.

PROFESSIONAL ACTIVITIES:

• Professional Development Workshop

1. **AI-Ready Educator Certificate (Center for Teaching & Learning, Kent State University):**
Recognized for completing a professional development program focused on understanding the impact of generative artificial intelligence in higher education and adopting responsible AI teaching practices. Participated in multiple AI-focused workshops and engaged in critical reflection on effective strategies to support students in navigating emerging AI technologies responsibly. Successfully fulfilled all program requirements and earned formal certification.
2. **Summer 2025 - Innovation Initiative: Artificial Intelligence (CTL, Kent State University):**
Selected as a cohort member for a competitive faculty development program focused on integrating generative artificial intelligence (GAI) into teaching, integrating AI policies in syllabus and instructional practice. Participated in a series of four workshops (July 22, 24, 29, and 31) centered on responsible AI use, classroom implementation, and pedagogical innovation. Completed all program activities and developed a final AI-enhanced instructional project to be shared with the academic unit and at a CTL-organized public event.
3. **Summer 2025 - Sensemaking Collaboratory:** Participated in a faculty development program designed to equip regional campus faculty with adaptive tools and leadership strategies. The program focused on navigating institutional complexity, fostering collaboration, and creating actionable tools that amplify the impact and visibility of regional universities within their communities.
4. **Fall 2024 - New Faculty Teaching & Mentoring Program (NFTMP):** A support program for new Kent State faculty across all campuses, offering mentorship, teaching development, and guidance on navigating Kent State's RTP (reappointment, tenure, promotion) and renewal processes.
5. **Summer 2024 - Belonging Changemakers Summer Project Program.** The project was selected to create opportunities for student engagement using Belonging principles in asynchronous courses.
6. **Spring 2024 - Belonging Champions Community of Transformation Workshop Series:** Participated in a faculty workshop series organized by the Center for Teaching and Learning (CTL), focused on fostering student success through promoting a sense of belonging. Received a stipend upon completion. The workshop covered evidence-based practices to enhance student belonging, real-time student feedback, and broader efforts to support student success in and beyond the classroom.
7. **Student Experience Project (SEP) Certificate** is a comprehensive training through the SEP Training Library. "Syllabus as a Tool to Promote Student Equity, Belonging, and Growth" Certificate ID: dt9q8zuods. Issued on December 28, 2023
Links:<https://library.studentexperienceproject.org/certificates/dt9q8zuods>
8. **Online Teaching Orientation: Basic Badge (2023):** I earned this certification after completing a rigorous program designed to enhance online teaching practices, focusing on creating interactive and inclusive digital learning spaces
9. **Spring 2023 - Faculty Anti-Bias Workshop Series:** A workshop series organized for faculty by the Center for Teaching and Learning (CTL) and the Division of Diversity, Equity, and Inclusion (DEI). Received a stipend on satisfactory completion of required reflections for each of the 5 sessions.
10. **Student Experience Project (SEP) Certificate** is a comprehensive training through the SEP Training Library. "Syllabus as a Tool to Promote Student Equity, Belonging, and Growth" **Certificate ID:** dt9q8zuods. Issued on December 28, 2023
Link: <https://library.studentexperienceproject.org/certificates/dt9q8zuods>
11. **Online Teaching Orientation: Basic Digital Badge** Earners of the Basic - Online Teaching Orientation & Refresher badge have completed a seven-module, fully online, asynchronous offering for Kent State faculty designed around essential online teaching topics and practices and supports faculty from all levels of teaching experience and environments. Issued on March 03, 2023
Link: <https://www.credly.com/badges/bb78bca6-16ba-4b69-9d0d-6096c8c85e4d>
12. Participation Certificate in **GPAD (Graduate Professional and Academic Development) workshop** series held at Kent State University, Kent, Ohio, USA on June 3, 2018.

13. Participation certificate of **IPC (Inter-Institutional Project Competition)** in 2013, India
14. Participation certificate in the sessions conducted by **Advait Life- Education** in 09.2010- 04.2011, India
15. Certificate for Completion On-line test conducted remotely by **IIT Bombay** for SciLab Workshop and Java Workshop in 2014, India

- **Professional Development & Engagement**

- **44th Original Lilly Conference on College Teaching (2025):** Participated in a national conference focused on evidence-based teaching and learning. Engaged in sessions on AI in education, Transparent Assignment Design (TILT), and innovative pedagogical practices in higher education.

- **Invited Talks**

- **Invited Speaker**, “*Retrieval-Augmented AI Feedback to Improve Metacognition and Equity in STEM Education*,” 2026 Faculty Idea Exchange (FIX), Columbus State Community College, Columbus, Ohio, USA, June 5, 2026. *50-minute invited faculty presentation.*
- **Invited Speaker**, “*Fostering Belonging in Asynchronous Learning: Building Inclusive, Engaging Online Courses*,” 2025 Ohio Celebration of Women in Computing (OCWiC’25), ACM-W, 2025. Received a **Speaker Badge of Recognition**.

- **Presentations and Panels**

- **Presenter**, “*EasyLearn: An AI-Powered Feedback System to Help Students Learn Better in STEM Courses*,” E2, Kent State University, 2025.
- **Presenter**, “*Belonging in Asynchronous Courses*,” OLC Innovate: Education Reimagined, Online Learning Consortium, 2025.
- **Presenter**, “*Fostering Belonging in Asynchronous Learning: Building Inclusive, Engaging Online Courses*,” E2, Kent State University, 2025.
- **Panelist**, “*Building Belonging in Asynchronous Environments: Roundtable Discussion on Strategies for Student Success and Well-Being*,” Innovative Teaching & Learning Conference, University of Tennessee, Knoxville, Tennessee, 2025.

- **Training:**

- **KSU Online Compliance Training (Vector Solutions) Certificate Issued on October 2024:**

1. **State and Federal Required Training:**

- * Fraud Reporting and Training

Certificate URL:

<https://kent-oh.safecolleges.com/...A6017E4-9285-11EF-A33C-61482C56E4A9>

2. **Beyond Compliance Training:**

- * Americans with Disabilities Act Overview

Certificate URL:

<https://kent-oh.safecolleges.com/...B1A27EE-928D-11EF-ADB2-0A4A62DB30FB>

- * Cybersecurity Overview

Certificate URL:

<https://kent-oh.safecolleges.com/...478FAF74-9295-11EF-B7BA-90381CE3692B>

- * Email and Messaging Safety

Certificate URL:

<https://kent-oh.safecolleges.com/...076A8F8A-9313-11EF-8882-76CEB2AFFBF1>

- * Implicit Bias and Microaggression Awareness
Certificate URL:
<https://kent-oh.safecolleges.com/...F3D55520-934B-11EF-B92A-5401220D9F38>
- * Making Schools Safe and Inclusive for Transgender Students
Certificate URL:
<https://kent-oh.safecolleges.com/...1DC8C818-935F-11EF-A32A-4D06220D9F38>
- * Managing Difficult Behaviors
Certificate URL:
<https://kent-oh.safecolleges.com/...671D4FC2-9362-11EF-98CA-2F4848470E1C>
- * Sexual Harassment: Staff-to-Staff
Certificate URL:
<https://kent-oh.safecolleges.com/...E4F75B5A-9316-11EF-8054-C702968AAEEC>
- Training on **Universal Axle Counter and Single Section Digital Axle Counter** from **Central Electronics Ltd.** in July 2012, India
- Training on **Embedded System** from **ATRIM Electronics Pvt. Ltd.** in July 2011, India
- Training on **Object Oriented Programming Using C++** from **NIIT** in August 2010, India

SERVICES:

University Service

- **Curriculum Committee Member, Stark Campus Representative**, Kent State University, Academic Year 2026–2027.
- **Stark Campus Representative (Non-Tenure Track Faculty)**, Kent State United Faculty Association (KSUFA), Kent State University, 2025–2026.
- *“Search Committee Member - IT Manager Hiring ” Served as a search committee member responsible for recruiting and selecting the IT Manager at Stark Campus, Kent State University.*
- **Outreach Workshop Instructor and Short-term School Organizations**
 1. *“Summer Enrichment Academy” (July 26-20, 2023)*
Instructor in the Summer Mathematics Enrichment Academy from July 26-20, 2023. Taught a module on “Scratch Me and Have Fun” for the students in the Data Science Academy. The academy was funded through the Mathematical Association of America’s prestigious Dolciani Mathematics Enrichment Grant.
 2. *“The 13th International Conference on Intelligent Human Computer Interaction ”*
(<https://www.ihci.cs.kent.edu/index.php/student-forum/>)
Visual Coding: (December 19, 2021): This workshop provides a quick introduction to programming with visual blocks like mBlock and Scratch.
- **Mentorship**
 1. *Research Mentor in Choose Ohio First Computer Science (COF-CS) for the 2023-26 Academic Years*
The CS-for-21st-Ohio program is a research-integrated scholarship initiative managed by the Department of Computer Science and supported by the State of Ohio’s ”Choose Ohio First Computer Science (COF-CS)” initiative.
<https://www.kent.edu/cs/global-circ/21st-century-computer-science-workplace-ohio>
 2. *Summer Undergraduate Research Experience (SURE) Mentor, Summer 2026, 2025 and 2024*
The SURE program provides funding for promising undergraduate researchers to engage in eight weeks of faculty-supervised research over the summer.
Accomplishment: Mentored an undergraduate student whose project on *Generative AI and Explainability* earned **1st Place** in the **Fall 2025 SURE 3-Minute Presentation Competition**.

3. *Project Mentor in RoboCup 2023*

RoboCup provides a platform to promote robotics and AI research through a publicly engaging but challenging competition. It represents a "grand challenge" project with significant social impact, aimed at advancing science and engineering.

<https://www.atr.cs.kent.edu/projects/robocup-2023>

– Judge for Research Presentations and Symposium (2023–2025)

1. *2026 Choose Ohio First Poster Symposium, April 19, 2026*
Cleveland State University, Cleveland, OH.
2. *Annual Undergraduate Symposium on Research, Scholarship and Creative Endeavors, April 13, 2026*
Kent State University, Kent, OH.
3. *2025 Choose Ohio First Poster Symposium, April 13, 2025*
Kent State University, Kent, OH.
4. *Annual Graduate Student Research Symposium, April 9, 2025*
Kent State University, Kent, OH.
5. *Summer Undergraduate Research Experience (SURE), October 21, 2024*
Kent State University, Kent, OH.
6. *17th Annual Stark Student Conference, April 19, 2024*
Stark Campus, OH.
7. *2024 Choose Ohio First Poster Symposium, April 7, 2024*
Case Western Reserve University, Cleveland, OH.
8. *Annual Graduate Student Research Symposium, April 1, 2024*
Kent State University, Kent, OH.
9. *2023 Choose Ohio First Poster Symposium, April 15, 2023*
Youngstown State University, Youngstown, OH.
10. *37th Annual Graduate Research Symposium, April 6–7, 2023*
Kent State University, Kent, OH.

Professional Services

Journal Paper Reviewer:

- **Frontiers in Computer Science – Human-Media Interaction (2026)**: Reviewed 1 paper.
- **Wiley – The Scientific World Journal (2026)**: Reviewed 1 paper.
- **Taylor & Francis – Computer Methods in Biomechanics and Biomedical Engineering: Imaging & Visualization (2025)**: Reviewed 1 paper.
- **MDPI Information (2026)**: Reviewed 1 paper.
- **MDPI Diagnostics (2026)**: Reviewed 1 paper.
- **MDPI Big Data and Cognitive Computing (BDCC) (2026)**: Reviewed 1 paper.
- **MDPI Electronics (2026)**: Reviewed 1 paper.
- **MDPI Sensors (2026)**: Reviewed 1 paper.
- **MDPI Symmetry (2026)**: Reviewed 1 paper.
- **MDPI Applied Sciences (2026)**: Reviewed 2 papers.
- **MDPI Acoustics (2026)**: Reviewed 1 paper.
- **MDPI Sustainability (2026)**: Reviewed 1 paper.
- **MDPI Journal of Imaging (2026)**: Reviewed 1 paper.
- **IEEE Transactions on Pattern Analysis and Machine Intelligence (2025)**: Reviewed 1 paper.

- **Springer Nature – Multimedia Tools and Applications (2025)**: Reviewed 1 paper.
- **MDPI Applied Sciences (2025)**: Reviewed 4 papers.
- **MDPI Sustainability (2025)**: Reviewed 3 papers.
- **MDPI Mathematics (2025)**: Reviewed 1 paper.
- **MDPI Journal of Clinical Medicine (2025)**: Reviewed 1 paper.
- **MDPI Symmetry (2025)**: Reviewed 1 paper.
- **MDPI Computers (2025)**: Reviewed 1 paper.
- **Springer Nature – Discover Computing (2025)**: Reviewed 1 papers.
- **MDPI BioMedInformatics (2025)**: Reviewed 1 paper.
- **Springer Nature – Artificial Intelligence Review (2026)**: Reviewed 4 papers.
- **Springer Nature – Archives of Computational Methods in Engineering (2026)**: Reviewed 3 papers.
- **Springer Nature – The Journal of Supercomputing (2026)**: Reviewed 1 paper.
- **Springer Nature – Scientific Reports (2026)**: Reviewed 2 papers.
- **Springer Nature – Discover Education (2026)**: Reviewed 1 paper.
- **Springer Nature – Humanities and Social Sciences Communications (2026)**: Reviewed 1 paper.
- **Springer Nature – Discover Artificial Intelligence (2026)**: Reviewed 1 paper.
- **Springer Nature – npj Artificial Intelligence (2025)**: Reviewed 3 papers.
- **Springer Nature – Archives of Computational Methods in Engineering (2025)**: Reviewed 1 paper.
- **Springer Nature – Discover Applied Sciences (2025)**: Reviewed 2 papers.
- **Springer Nature – BMC Medical Imaging (2025)**: Reviewed 2 papers.
- **Springer Nature – Discover Data (2025)**: Reviewed 1 paper.
- **Springer Nature – Neural Processing Letters (2025)**: Reviewed 2 papers.
- **Springer Nature – AI and Ethics (2025)**: Reviewed 1 paper.
- **PeerJ Computer Science (2025)**: Reviewed 3 papers.
- **Frontiers in Medicine – Pathology (2025)**: Reviewed 1 paper.
- **Cybernetics and Systems, Taylor & Francis (2025)**: Reviewed 1 paper.
- **International Journal of Grid and Utility Computing (IJGUC) (2025)**: Reviewed 1 paper.
- **International Journal of Pervasive Computing and Communications (IJPCC) (2025)**: Reviewed 1 paper.
- **Engineering, Technology & Applied Science Research (ETASR) (2025)**: Reviewed 1 paper.
- **IEEE Open Journal of the Computer Society (OJ-CS) (2024)**: Reviewed 1 paper.

Conference Paper Reviewer:

- **IEEEVIS - 2026**: Reviewed 1 paper.
- **EAAI - 2026**: Reviewed 4 paper.
- **ACM CHI-2026**: Reviewed 1 paper.
- **IEEE CCWC-2026**: Reviewed 4 paper.
- **IEEE UEMCON-2026**: Reviewed 2 papers.
- **IEEE AIIoT- 2025**: Reviewed 4 paper.
- **IEEE ACDSA- 2025**: Reviewed 1 paper.

- **IEEE CCWC-2025**: Reviewed 5 paper.
- **IEEE ICPCT-2025**: Reviewed 3 paper.
- **IEEE ICECCME 2024**: Reviewed 1 paper.
- **ICDSNE 2023**: Reviewed 4 paper.
- **MobileHCI 2023**: Reviewed 1 paper.

PRESENTATION:

1. *Presentation at the 2026 IEEE Conference on Artificial Intelligence (IEEE CAI 2026) on “Retrieval-Augmented AI Feedback to Support Metacognition and Accessible Feedback in STEM Education”, May 8–10, 2026, Granada, Spain.*
2. *Presentation at the 2026 IEEE Conference on Artificial Intelligence (IEEE CAI 2026) on “Interpreting Text-to-Image Diffusion Through Structured Prompts and Cross-Attention Attribution”, May 8–10, 2026, Granada, Spain.*
3. *Presentation at the Visualization and Data Analysis (VDA 2026) conference on “Easy Interpretation of Image Classification Results with Feature-Level Visualization”, January 2026, San Francisco, USA.*
4. *12th International Conference on Soft Computing & Machine Intelligence (ISCMi 2025) on “A Survey of DeepSeek Models”, November 21–23, 2025, Rio de Janeiro, Brazil.*
5. *7th IEEE International Workshop on Artificial Intelligence and Education (WAIE 2025) on “DeepSeek Models in STEM Education: Capabilities, Applications, and Challenges” September 27-29, 2025 (Online Presentation), Yokohama, Japan*
6. *34th IEEE International Conference on Computer Theory and Applications (ICCTA 2024), “A Tiered GAN Approach for Monet-Style Image Generation” at the ICCTA 2024 conference, Alex, Egypt, held from December 14-16, 2024 (Online Presentation).*
7. *IEEE International Conference on Knowledge Innovation and Invention (ICKII) on “Deepening Mathematical Understanding Using Visualization and Interactive Learning for Deaf Students” 16 - 18 August 2024 (Online Presentation), Nagoya, Aichi, Japan*
8. *2024 IEEE World AI IoT Congress (AIIoT) on “Exploring Fine-Grained Feature Analysis for Bird Species Classification using Layer-wise Relevance Propagation” 29 - 31 May 2024, Seattle Convention Center, Seattle, WA, USA*
9. *27th International Conference Information Visualisation (IV) on “BookMate: Leveraging Deep Learning to Empower Caregivers of People with ASD in Generation of Social Stories.” 25 - 28 July 2023, Tampere University, Tampere, Finland*
10. *13th International Conference on Intelligent Human-Computer Interaction (IHCI-2021) “Interactive Visualization and Capturing Geo-Coded Multimedia Data on Mobile Devices” 20 -22 December, 2021, Kent State University, Kent, Ohio, USA*
11. *“Mobile based Visualization Techniques and Tools for Storytelling”*
[\(https://www.ihci.cs.kent.edu/index.php/student-forum/\)](https://www.ihci.cs.kent.edu/index.php/student-forum/)
 Student Forum, 13th International Conference on Intelligent Human-Computer Interaction (IHCI-2021) December, 2021
12. *“Embedded System”*
 Blitz Computer Society (First Prize Winner)
 July 2012.

SOCIAL SERVICES AND LEADERSHIP:

1. **Faculty Co-Advisor**, Stark Computer Club, 2022 to Present
2. **Membership**, IEEE and ACM, 2022 to Present
3. **Vice President**, ACM-W, Computer Science Kent State University, Ohio, USA, 2020 to 2022
4. **Member**, University Diversity Action Council (UDAC), USA, 2022 to 2025: Involvement began with training as a new member, progressing to my current role on the consultation committee, where I serve as a co-lead for the Women's Center.
5. Kent State University Silver Sponsor representative at **OCWiC ACM-W Professional Chapter 2023**
6. **Technical Program Committee (TPC)**: TPC member for the IEEE World AI IoT Congress 2025, contributing to the review process and selection of high-quality research submissions for the conference.
<https://worldaiiotcongress.org/com/technical-committee/>
7. **Technical Program Committee (TPC)::** TPC member for the International Conference on Pervasive Computational Technologies (ICPCT-2025), contributing to the review process and selection of high-quality research submissions for the conference.
<https://www.glbitm.org/icpct-2025/TPC.html>
8. **Event Logistic Leadership**, 13th International Conference on Intelligent Human-Computer Interaction (IHCI-2021) held at Kent State University, Kent, Ohio, USA from 12.19.2021 to 12.22.2021
9. **Volunteer in IHCI-2021 Conference** held in Kent State University, Kent, Ohio, USA on 12.20.2021
10. **Member**, Computer Science Graduate Student Association, KSU, Ohio, USA, 2018 to 2022
11. **Treasurer**, International Graduate Student Association, KSU, Ohio, USA, 2017 to 2020
12. **Event Organizer and Treasurer**, National Service Scheme (NSS) camp program 08.2010 to 03.2013
13. **Event Organizer**, Blood Donation Camp organized by Rotary Ghaziabad North, Uttar Pradesh, India on 2012 to 2013
14. **Event Planner and Treasurer**, College Cultural Event, India from 2012 to 2013